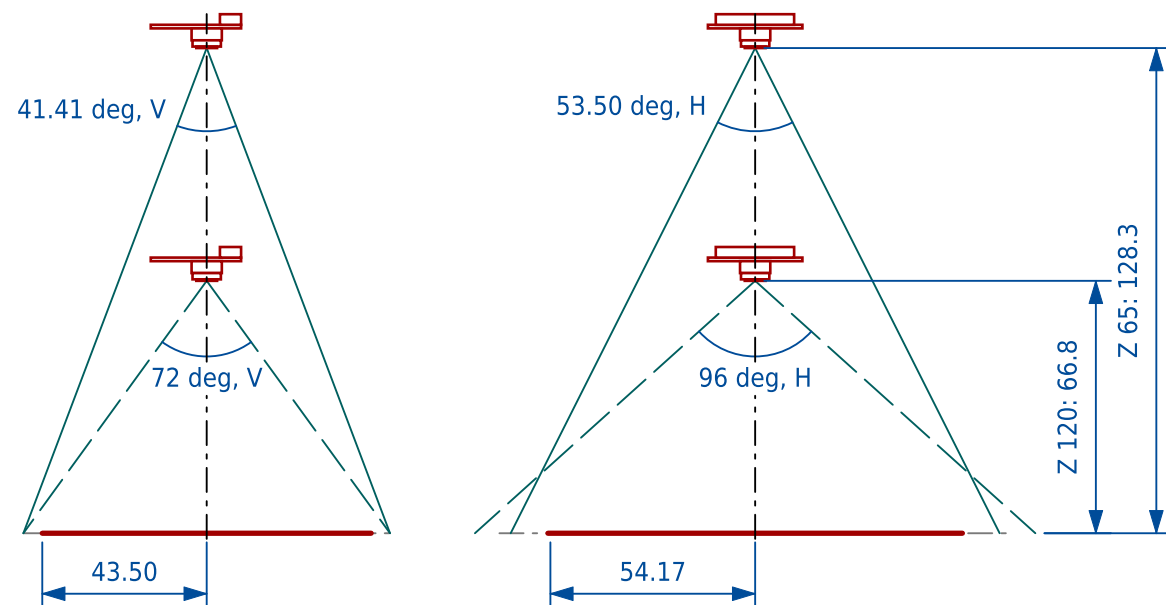


END ELEVATION

FRONT ELEVATION

**NOTES**

1. First angle; the end elevation is seen from the right. The camera, a Camera Module v1.3 from its own data, drawn per lens and fitted one at a time, looks straight down, in a mode reading the whole sensor. Z is to its entrance pupil, which nobody locates; ASSUMED behind the lens face by at most the lens's 5.20 mm, so set the FACE at Z and the picture is up to 7.8% larger, never smaller.

FRAMES, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y
A	Whole board	129.33 x 97.00	X	54.17	43.50
B	LD0-LD7	32.68 x 24.51	X	19.50	6.83

Z ABOVE THE FRAME PLANE, mm, AND IN FOCUS THERE?

	Z 65	FIXED	Z AF	MOTOR	Z 120	FIXED	MOTOR
A	128.3	no	129.7	yes	66.8	no	none listed
B	32.4	no	32.8	no	16.9	no	none listed

LENSES, deg, AS USED: SEE RPICAM-LENS

	MODULE	H	V	DIAG	f mm	F	FOCUS FROM
65	RPi Camera v1.3	53.50	41.41	64.4 DERI	3.60 DECL	F2.9 DECL	Fixed, 1 m
AF	Arducam B0176	54	41	64.6 DERI	3.67 DERI	F2.9 ASSU	Motor, 80 mm
120	Arducam B006604	96	72	120.0 DECL	2.17 DERI	F2.4 ASSU	Fixed, 1 m

LEGEND

- Subject outline
- LED, display or connector
- - - Adjacent part or connector body
- Frame A's target, on the plane Z is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- The 120 deg lens's field of view, 96 x 72, to its picture's edge
- Frame A, the rectangle the picture must cover
- Frame B, the rectangle the picture must cover
- Dimension, extension and leader

NOTES (continued)

2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round, turned whichever way needs the lower camera; LONG says which. One margin absorbs both where the stand ends up -- 5.00 mm of lateral error, or a lean of 2.0 deg at frame A's 65 Z or 2.9 at its 120, measured at the picture's edge -- and what is not known about the lens, below: what one uses the other cannot.
3. PLANE, which Z is measured from: A/B from the subject's own top face. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. FOCUS. Both fixed lenses are declared "Approx 1 m to infinity" (Raspberry Pi) and "Focus Distance 1 m to infinity" (Arducam), and every Z here is nearer: on either the board is OUT OF FOCUS. At frame A's Z a point spreads to 23 px, 1.1 mm on the board, at 65, and 20 px, 0.8 mm, at 120 (DERIVED, thin lens ASSUMED set at 2 m). The autofocus AF, Arducam's B0176, is declared "80mm to infinity": every Z of 80 or more is in focus once it has focused; at frame A's its depth of field is 120.6 to 140.4 (DERIVED, F2.9 ASSUMED). Arducam's catalogue lists no 120 deg OV5647 with a motorised lens.

NOTES (continued)

5. The 120 deg lens is a fisheye, 120 deg on the diagonal: 96 x 72 is Arducam's split of that diagonal, equidistant, not a measurement. Its barrel distortion bows the picture's edges outwards on the board, so the frame's corners are inside it; the margin absorbs the narrower lenses the evidence allows, 120 diagonal, equisolid, 94.3 x 69.8, leaving 3.1 mm; Commonlands CIL282, scaled to 120, 94.4 x 70.8, leaving 4.0 mm; YXF4Y001A1, 1.79 mm, 92.4 x 73.9, leaving 6.2 mm. See RPICAM-LENS.
6. The "65 deg" name is the stock lens's DIAGONAL, which no vendor prints: $2 \times \arctan(\sqrt{3.6736^2 + 2.7384^2} / 2 / 3.60) = 64.94$ deg on OmniVision's image area, 64.42 on the pixel array. Every Z here is from the angle along each axis. 65 deg across frame A's long side would give Z 101.5 and lose 13.5 mm off each end.
7. Quotes are transliterated to ASCII: x, um and infinity for the signs the cached pages print.
8. Frame A, Whole board: The assembled envelope: the outline together with the connectors and Pmod bodies that overhang it.
9. Frame B, LD0-LD7: The four tri-colour LEDs LD0-LD3 and the four single LD4-LD7, in two rows on a 7.00 mm pitch.

NOTES (continued)

10. Which LED row is which is not named by any Digilent source; FPGA-ARTY-A7 takes the row nearest the edge as the tri-colour LD0-LD3 by package size. Either way both rows are inside the frame, so the framing does not turn on it.

SOURCES

1. Board geometry: fpga/boards.py - Outline, Pmod hosts, connectors and LED rows; see FPGA-ARTY-A7.
2. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE ARTY A7

Camera Module OV5647, 65 and 120 degree lenses

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-352-g967eb38	mm	1:2, PLAN 1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-OVER-ARTY	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
Digilent Arty A7	DXF +/-0.20, plot bodies +/-0.30, Z DERIVED		